C12.4 Physical and virtual address space size

The DVM protocol can be extended to support different combinations of virtual and physical address space sizes. For any component, any of the following can be true:

- physical address space size matches virtual address space size
- physical address space size exceeds virtual address space size
- virtual address space size exceeds physical address space size.

C12.4.1 Physical address space size matches virtual address space size

In common usage, a component has a matching physical address space size and virtual address space size. Address information is transferred using **ARADDR** with no adaptation or special considerations required.

C12.4.2 Physical address space size exceeds virtual address space size

If a component supports a larger physical address space than virtual address space, the number of bits in the **ARADDR** signal matches that of the physical address space size and no special considerations are required for physical address based operations.

Virtual address operations might receive additional address information in a DVM transaction. However, it is a DVM protocol requirement that any additional address information is ignored and operations are performed using only the supported address bus bits. In certain circumstances this approach can result in an over-invalidation, where more entries are invalidated in the cache than is functionally required. However this is acceptable for read-only structures and is functionally correct.

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C12.4.3 Virtual address space exceeds physical address space

If a component supports a larger virtual address space than its physical address space, then a minimum **ARADDR** payload size is required to support the virtual address space size, as Table C12-1 shows. This then requires that the component takes appropriate action with regard to the additional physical address bits.

Table C12-1 Supporting virtual address spaces larger than physical address spaces

ARADDR width required	Maximum virtual address space size	ARADDR bits
32-bit	32-bit	Uses the following pins for virtual address information: **1st transaction** **[31:24]** *Virtual Machine Identifier* (VMID) **[23:16]** *Application Space identifier* (ASID) **2nd transaction** **[31:4]** Virtual address bits [31:4]
40-bit	48-bit	Uses the following pins for virtual address information: **1st transaction** **[39:32]** Virtual address bits [47:40] **[31:24]** VMID **[23:16]** ASID **2nd transaction** **[39:4]** Virtual address bits [39:4]
48-bit	64-bit	Uses the following pins for virtual address information: **1st transaction** **[47:32]** Virtual address bits [63:48] **[31:24]** VMID **[23:16]** ASID **2nd transaction** **[47:4]** Virtual address bits [47:4]

Interoperability considerations on page C6-239 describes the general rules for the interaction of two components with different physical address space sizes. These rules must be applied to any component that has a wider address bus than its naturally supported physical address space size.

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